Homework 8
18 March 2025
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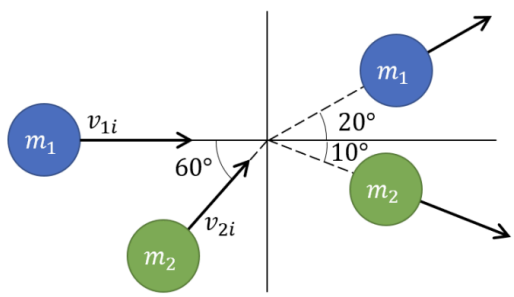
Question 1

Given:
Two balls colliding while rolling on a smooth surface.
Ball 1 has a mass $m1$ of 4.00 kg and travels with an initial velocity of 2.55 m/s.
Ball 2 has a mass $m2$ of 2.61 kg and travels with an initial velocity of 1.55 m/s.

Find:
After the collision, what is the velocity of (a) ball 1 and (b) ball 2?

Diagram:



Theory:
$u_n = v_f$ of ball n

$$u_1 = \frac{(m_1 - m_2)v_1 + 2m_2v_2}{m_1 + m_2}$$

$$u_2 = \frac{(m_2 - m_1)v_2 + 2m_1v_1}{m_1 + m_2}$$

Assumptions:
We assume a perfectly elastic collision and no other forces acting on the isolated system.

Solution:

$$u_1 = \frac{(4-2.61)(2.55)+2(2.61)(1.55)}{4+2.61} = \boxed{1.76\ m/s}$$

$$u_2 = \frac{(2.61-4)(1.55)+2(4)(2.55)}{4+2.61} = \boxed{2.76\ m/s}$$

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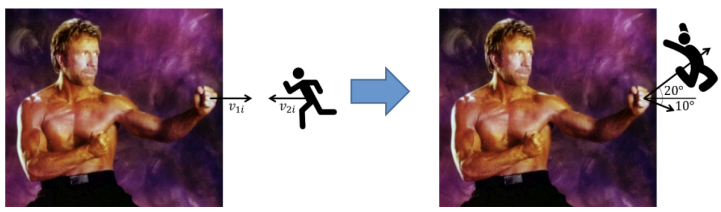
Question 2

Given:
A villain with mass 84.0 *kg* runs at a velocity of 9.26 *m*/*s* directly into Chuck Norris' fist moving at a speed of 0.150 *m*/*s*. The villain is knocked back at a velocity of 8.52 *m*/*s* and angle 20.0° from the horizontal, while Chuck Norris' fist moves at a velocity of 0.0745 *m*/*s* and 10.0° from the horizontal.

Find:
a) What is the mass of Chuck Norris' fist in *kg*?
b) How much kinetic energy is lost in the collision?

Diagram:



Theory:
Momentums in the X and Y directions respectively.

$$m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} \cos(10°) + m_2 v_{2f} \cos(20°)$$

$$0 = -m_1 v_{1f} \sin(10°) + m_2 v_{2f} \sin(20°)$$

$$K_T = \frac{1}{2}[\, m_1 (v_{1T})^2 + m_2 (v_{2T})^2 \,] \;\; ; \;\; T \in \{i, f\}$$

Assumptions:
We assume a perfectly inelastic collision and no other forces acting on the isolated system.

Solution:

a) $m_1 = \frac{(84.0)(8.52)(\sin(20°)=0.3420)}{(0.0745)(\sin(10°)=0.1736)} = \boxed{18950\ kg}$

b) $K_i = \frac{1}{2}[\,(18950)(0.150)^2 + (84.0)(9.26)^2\,] = 3813\ J$

$K_f = \frac{1}{2}[\,(18950)(0.0745)^2 + (84.0)(8.52)^2\,] = 3102\ J$

$\Delta K = |K_f - K_i| = |3102 - 3813| = \boxed{711\ J}$

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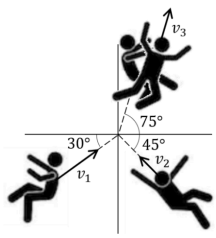
Question 3

Given:
Ender and Shen are flying at each other during a battle in space. Ender moves with a velocity $v1$ of 9.660 m/s while Shen, with a mass of 53.8 kg, moves at a velocity $v2$ of 7.26 m/s. When the two collide, they hold on to each other and move with a velocity $v3$ of 5.164 m/s.

Find:
What is Ender's mass, in kg?

Diagram:



Theory:
$m_1 v_1 + m_2 v_2 = (m_1 + m_2) \cdot v_3$

Assumptions:
We assume a perfectly inelastic collision and no other forces acting on the isolated system.

Solution:

$m_1 = \frac{m_2(v_3 - v_2)}{v_1 - v_3}$ rearrange to solve for m_1

$m_1 = \frac{(58.8)(5.164-(-7.26))}{9.660-5.164} = \boxed{148.7\ kg}$